

SOS SIGNALS

FLARES

During an emergency at sea or on a mountain, it is important for rescuers to be able to locate those in danger as quickly as possible. One way to signal for help is by using flares. These contain chemicals that burn with a bright, colored flame that can be seen from miles away, pinpointing where the rescuers need to go. They can come in different colors, which are produced by different chemicals, but most are a deep red. Flares can be used both during the day and at night. Some flares emit thick smoke instead of light. These are used in places where the ground is not clearly visible from the sky, such as forests.

GUNPOWDER

Flares contain a chemical that releases oxygen when heated, making the flare burn brightly. Gunpowder also contains this chemical, along with sulfur and charcoal. While flares are designed to burn safely, gunpowder is tightly packed to explode.



FLAME TESTS

When put into a hot flame, different metal salts cause the flame to change color. Scientists can use these "flame tests" to work out which metal is present in a particular salt. A wire loop dipped in a small amount of the salt is placed in a blue Bunsen flame, and the color is observed. Bright red means strontium is present, while calcium gives orange and sodium gives yellow flames.

